25901387 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMAT203 - MATHEMATICAL TOOLS FOR COMPUTING

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. What is the least common multiple of $2^3.3^5.7^2$ and $2^4.3^3.5$? [U] / [CO1]
2. Using the hashing function h(k)=kmod13, find the memory locations assigned to the following keys: 45, 56. [A] / [CO1]
3. Draw an OR gate with 'n' inputs [A] / [CO2]
4. Find the duals of a) $\overline{(x.1)} + \bar{y}.0$ b) $xy + \bar{x}\,\bar{y}$ [U] / [CO2]
5. Let A be the set {1, 2, 3, 4}. Which ordered pairs are in the relation R = {(a, b) | a divides b }? [U] / [CO3]
6. Let R1 and R2 be relations from A = {1, 2, 3} to B = {1, 2, 3, 4} such that R1 = {(1, 1), (2, 2), (3, 3)} and R2 = {(1, 1), (1, 2), (1, 3), (1, 4)} .Then find R1 ∪ R2 and R1 ∩ R2 [U] / [CO3]
7. Draw the Binary Search Tree (BST) for the data: 20, 16, 15, 25, 60, 40, 30 [A] / [CO4]
8. What is meant by a weighted graph?. Give an example [K] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Find the value of (177 mod 31 . 270 mod 31) mod 31 [A] / [CO1]

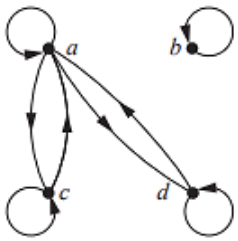
10. Express gcd(252 , 198) = 18 as a linear combination of 252 and 198 . [A] / [CO1]

11. a) Find the value of $(1.1) + \overline{(1.1) + (0.1)}$ [An] / [CO2]
    b) Translate the logical equivalence (T ∧ T) ∨ ¬F ≡ T into an identity in Boolean algebra

12. a) Write all the minterms of a 2-variable Boolean function. [U] / [CO2]
    b) Find the Boolean expression that represent the function F(x,y) given in the following table.

x	y	F(x,y)
1	1	1
1	0	0
0	1	1
0	0	0

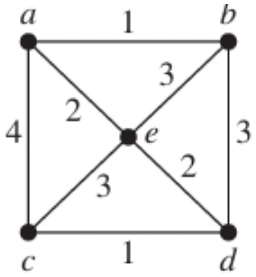
13. Determine whether the relation represented by the given directed graph is an equivalence relation. [U] / [CO3]



14. Draw the directed graph representing the relation R on the set A= {1,2,3}, where R= {(1, 2), (1, 3), (2, 3), (3, 2), (3,3)}. Also, write the matrix of the relation. [U] / [CO3]

15. Describe the **Depth-First Search (DFS) algorithm**. Illustrate with a graph. [A] / [CO4]

16. Use Prim's algorithm to find a minimum spanning tree for the following weighted graph. What is its weight? [A] / [CO4]



[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Encrypt the message " STOP POLLUTION " using the encryption function f (p) = (17p + 22) mod 26 [A] / [CO1]

18. State and prove the distributive laws in Boolean algebra. [An] / [CO2]

19. Let R1 and R2 be relations on a set A represented by the matrices [A] / [CO3]

$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find the matrices that represent

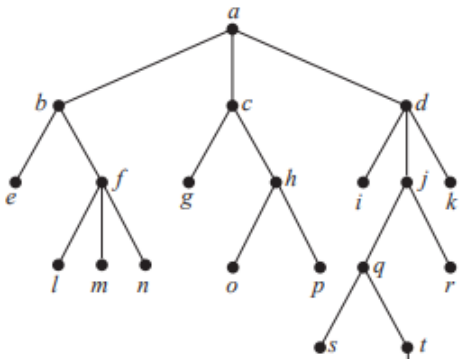
a) R1 ∪ R2. b) R1 ∩ R2. c) R2 ◦ R1. d) R1 ◦ R1. e) R1 ⊕ R2.

20. a). Define partial ordering. Show that the divisibility relation '|' is a partial ordering on the set of positive integers [A] / [CO3]

b). Draw the Hasse diagram representing the partial ordering R= {(a, b)|a divides b} on the set A= {1, 2, 3, 4, 6, 8, 12, 24}.

21. a) Briefly explain the preorder, inorder, and postorder traversals of a rooted tree. [A] / [CO4]

b) Find the preorder, inorder, and postorder traversals of the following tree.



22. Explain rooted trees in detail. Define and explain parent, child, siblings, ancestors, descendants, internal vertices, leaves, and subtrees with a diagram. [U] / [CO4]

[10x3 = 30]